



BorNEO HackWknd 2026

REPORT

Title of Project:	MySemai AI
Team's Track:	Technical
Case Study chosen:	2. Indigenous Language Preservation
Team Name:	EchoLingo
Group Leader:	Chong Jing
Group Members:	Teoh Jing Ling Teh Xin Choh Zhen Xuan

Table of Content

1.0 Introduction.....	3
1.1 Background of the Case Study.....	3
1.2 Chosen Sustainable Development Goal (SDG).....	3
1.3 Problem Statement.....	3
1.4 Objectives.....	4
2.0 Review of Existing Solution.....	5
2.1 Existing Language Preservation Platforms.....	5
2.2 Limitations that Aims to Fix.....	5
3.0 Methodology.....	7
3.1 Technology Stack.....	7
3.2 System Architecture.....	7
3.3 Implementation Process.....	8
4.0 Result.....	9
5.0 Prototype Showcase.....	11
5.1 Landing Page.....	11
5.2 AI Lip Reading Feature.....	11
5.3 Lip Reading Recognition Result.....	12
5.4 Vocabulary Library.....	12
5.5 Save New Word Feature.....	13
5.6 MySemai AI Chatbot.....	13
5.7 Cultural Information Page.....	14
6.0 AI Acknowledgement.....	15
7.0 Conclusion.....	15
Reference.....	17

1.0 Introduction

1.1 Background of the Case Study

Southeast Asia, a region that is famous for its rich cultural diversity which consists of hundreds indigenous languages that represent the identity, history and traditions of different communities. These languages were not only just a way of communication but also carried the stories, beliefs and knowledge that passed down from the elders to the younger generations. However, these languages are gradually disappearing nowadays due to globalization, migration to urban areas and dominance of national and international languages in schools, social media and technology. UNESCO states that without serious preservation efforts, many of these indigenous languages may eventually become extinct.

In Malaysia, the Semai community is one of the largest Orang Asli groups in Peninsular Malaysia. They speak Bahasa Semai which is a language that is passed down orally. Challenges such as limited digital documentation, lack of education resources and minimal technological support causes a decline in the use of Bahasa Semai among the younger generations. Most Artificial Intelligence (AI) and Natural Language Processing (NLP) systems focus on widely spoken languages such as English and Mandarin which further led to Bahasa Semai being underrepresented in the digital world.

With the rapid growth of AI technology, digital platforms contribute an opportunity to support language revitalizations. Interactive learning tools, speech recognition and digital archives can help preserve and promote Bahasa Semai ensuring it remains relevant and accessible for future generations.

1.2 Chosen Sustainable Development Goal (SDG)

This project aligns with United Nations Sustainable Development Goal 4: Quality Education. SDG 4 emphasizes inclusive and equitable quality education and promotes lifelong learning opportunities for all. By providing this accessible, AI-powered learning platform for Bahasa Semai supports inclusive digital education for indigenous communities and promotes linguistic diversity.

Additionally, the project indirectly contributes to SDG 10: Reduced Inequalities by addressing digital language inequality and SDG 11: Sustainable Cities and Communities by safeguarding cultural heritage. However, the primary focus remains on SDG 4 for this project.

1.3 Problem Statement

Indigenous languages such as Bahasa Semai are at risk of gradual extinction due to limited digital presence and documentation as well as lack of structured online learning platforms. In addition, the absence of AI-based pronunciation and speech learning tools and reduced usage

among younger generations in digital environments led to indigenous languages being underrepresented nowadays.

Most existing online language platforms focus on widely spoken languages and only offer a little support for minority or indigenous languages. Furthermore, traditional online language libraries such as Wikipedia are mostly text- and audio-based which lack interactive pronunciation feedback mechanisms. Due to this limitation, learners have fewer opportunities to practice and improve their speaking skills. Without innovative digital solutions, Bahasa Semai may gradually become invisible in the modern digital landscape.

Hence, there is a need for an AI-powered and interactive platform that not only archives Bahasa Semai but also actively supports language learning and pronunciation practice using advanced technologies such as lip-reading, speech analysis and AI-assisted learning tools.

1.4 Objectives

The main objectives of this project are:

1. To develop an AI-powered web-based learning platform that focuses on Bahasa Semai language preservation and education.
2. To create a dynamic digital language library that stores vocabulary, meanings, audio pronunciations, and cultural information in an organized and expandable format.
3. To integrate an AI-based lip-reading system that analyzes users' mouth movements through camera and provides real-time feedback to improve their pronunciation.
4. To encourage community participation by allowing native speakers to contribute new words and pronunciation recordings.
5. Integrate an AI chatbot assistant to help users learn and understand Bahasa Semai vocabulary.
6. To promote inclusive digital education for indigenous communities in alignment with SDG 4 (Quality Education).

Through these objectives, the project aims to narrow the gap between traditional oral knowledge and modern digital technology, ensuring that Bahasa Semai remains relevant and accessible in the AI-driven era.

2.0 Review of Existing Solution

2.1 Existing Language Preservation Platforms

Several digital platforms and applications have been developed to preserve and promote endangered or indigenous languages. These solutions usually focus on creating digital archives, providing language learning resources, or enabling communities to document their languages.

One example is FirstVoices, it is a web-based platform that developed to support the Indigenous communities in documenting and promoting their languages. The platform allows users to upload vocabulary, audio pronunciations, and cultural information that is related to their language. It acts as a digital archive where users can explore language materials and learn basic vocabulary and pronunciation.

In addition, another example is Aikuma which is a mobile application designed to collect speech recordings and translations from native speakers. This app is particularly essential for documenting endangered languages in remote communities because it allows researchers and community members to record spoken language and align it with translations for linguistic documentation.

There are also educational language-learning applications such as IndyLan. This platform provides structured lessons that include vocabulary, phrases, dialogues, and cultural content for several endangered European languages. It also includes audio pronunciations and interactive learning modes that help learners practice vocabulary and improve listening comprehension.

Moreover, some innovative tools use artificial intelligence for language education. For instance, Woolaroo uses machine learning and image recognition technology to identify objects through a smartphone camera and display their names in indigenous languages along with pronunciation guidance.

Therefore, these platforms demonstrate how digital technology can support the revitalization and learning of endangered languages by providing online resources and interactive learning environments.

2.2 Limitations that Aims to Fix

Although existing platforms act as a significant role in language preservation, several limitations still remain.

First, many platforms mainly focus on language documentation rather than interactive learning environments. For example, digital archives like FirstVoices focus on storing vocabulary and recordings, but they offer limited tools for users to actively practice pronunciation or improve their speaking skills.

Second, many language-learning applications rely on text and audio materials. While audio recordings help users to hear the correct pronunciation, they do not provide feedback on whether they are pronouncing the word correctly.

Third, advanced AI technologies are rarely used in indigenous language learning platforms. Features such as real-time pronunciation feedback, facial recognition, or lip movement analysis are generally absent in most existing systems.

Furthermore, another limitation is that most platforms are not designed for continuous community contribution. As indigenous languages change and grow over time, the digital language resources may become outdated without a collaborative platform where native speakers can easily add new vocabulary, stories, and recordings.

Finally, many global language-learning platforms prioritize widely spoken languages, which leaves minority languages with limited technological support and fewer digital learning tools.

3.0 Methodology

3.1 Technology Stack

The development of this project uses several modern web technologies to create an interactive and AI-powered platform for learning Bahasa Semai. The frontend of the platform is built using standard web technologies such as HTML, CSS, and JavaScript to design a responsive and user-friendly interface. These technologies allow users to navigate smoothly between different sections of the platform which includes the landing page, lip-reading scanner, vocabulary library, word contribution page, and the MySemai AI chatbot without needing to refresh the browser page.

Artificial intelligence services are integrated to support the main interactive features of the platform. The lip-reading feature uses the device camera to capture the user's lip movements and analyze them using an AI recognition model to identify the spoken Bahasa Semai word. After the analysis is completed, the system will display the predicted word together with its English meaning and a confidence score that indicates how accurate the prediction is.

In addition, the platform includes an AI chatbot called MySemai AI, which is connected to Google Gemini. The chatbot allows users to ask questions related to Bahasa Semai vocabulary and receive clear explanations or learning assistance in real time. This feature helps users to gain a better understanding of the language while interacting with the platform.

For data storage, a cloud-based database is developed to store vocabulary information, user-contributed words, and recorded pronunciation clips. This database enables the platform to maintain a growing language library as users continue to contribute new vocabulary and recordings.

3.2 System Architecture

The system follows a client-server architecture where users interact with the web platform through a browser-based interface. The frontend interface manages user interactions such as browsing the vocabulary library, activating the lip-reading scanner, contributing new words, and communicating with the AI chatbot.

When a user activates the lip-reading scanner, the system will capture short video clips of the user's lip movements through the camera. These clips are then sent to the AI recognition system and analyzes the lip movements to identify the most likely Bahasa Semai word. The system then returns the predicted word along with its English meaning and a confidence score to demonstrate the reliability of the recognition result.

The platform also includes a vocabulary database that stores all Bahasa Semai words, their meanings, and pronunciation recordings. Users can contribute new vocabulary through the "Save

New Word” feature by entering the new Bahasa Semai word and its meaning. The system then guides the user to record multiple short pronunciation clips and store them in the database to help build a dataset for improving the AI recognition system.

In addition, the MySemai AI chatbot connects with the Google Gemini model to generate instant responses. This allows users to ask questions about Bahasa Semai words and receive helpful explanations directly within the platform.

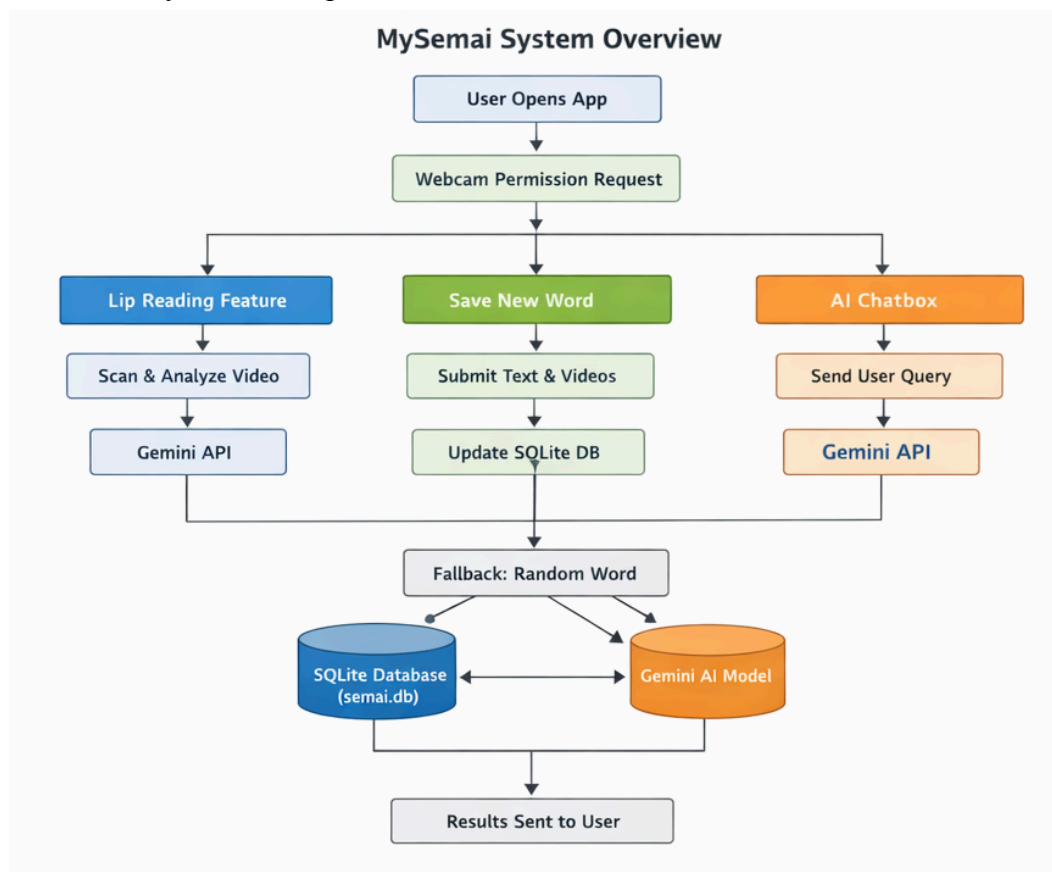


Figure 3.2: System Architecture and Workflow of the MySemai AI Platform

3.3 Implementation Process

The implementation of the system was carried out in several stages. First, the web interface was designed to provide a simple and intuitive navigation structure. The platform was implemented as a single-page application to allow users to move between different sections smoothly without experiencing page reloads.

Next, the lip-reading scanner function was integrated into the system by connecting the device camera to the AI recognition model. The system continuously records short video clips of the user’s lip movements and sends them to the AI model for analysis. The model then predicts the most likely Bahasa Semai word and displays the result together with its confidence score.

After that, the vocabulary library and contribution system were developed. Users can browse the stored vocabulary through the library page, while the “Save New Word” feature allows users to contribute new words and record pronunciation clips. These contributions are automatically stored in the database and displayed in the vocabulary library.

Finally, the MySemai AI chatbot was integrated into the platform by connecting the system to the Google Gemini API. This allows users to interact with an intelligent assistant that can answer questions about Bahasa Semai vocabulary and provide additional learning support.

Through this implementation process, the platform combines web technologies and artificial intelligence to create an interactive learning environment that supports the preservation and learning of Bahasa Semai.

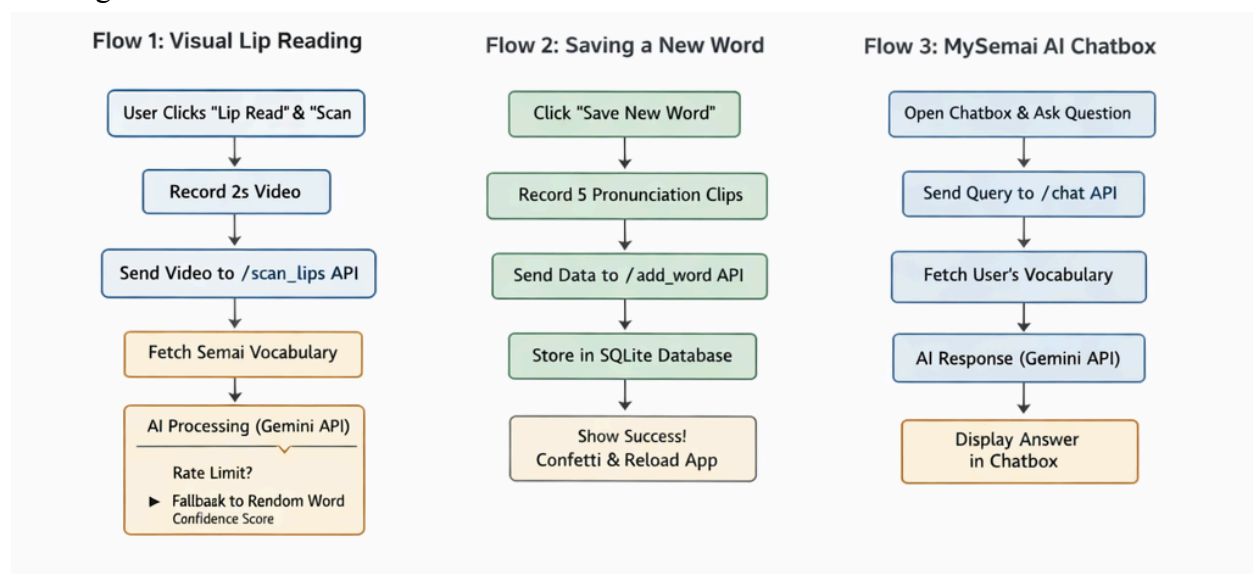


Figure 3.3: Workflow of Key Features in the MySemai AI Platform, including the visual lip-reading recognition process, new vocabulary contribution process, and MySemai AI chatbot interaction.

4.0 Result

The project resulted in a functional prototype of a web-based platform designed to support the learning and preservation of Bahasa Semai. The platform successfully integrates several key features, such as digital vocabulary library, AI-based lip-reading pronunciation system, crowdsourced data collection function and AI chatbot assistant. These features work together to create an interactive and engaging education environment for users who want to learn Bahasa Semai.

The main objective of this project was to create an easily accessible platform for users to learn and preserve Bahasa Semai through digital technology. Based on the platform developed, users

are able to browse Bahasa Semai digital vocabulary library, practice pronunciation by using camera-based lip-reading feature and contribute new words to expand the language database. Hence, this demonstrates that the main mission of this project has been achieved at the prototype stage.

By clicking the “Lip Reading” feature of the platform, the AI-based lip-reading feature allows the system to capture short video clips of the user’s lip movement through the camera. The AI recognition model analyzes the lip movement and identifies the most likely Bahasa Semai word. The recognised word is then displayed together with its English meaning and a confidence score that indicates the accuracy of the user's pronunciation. This interactive function encourages users to actively practice their speaking skills instead of only reading the vocabulary.

In addition, the platform allows users specially the native speaker to contribute new Bahasa Semai words through the “Save New Word” feature. Users are able to key in the new Bahasa Semai word, add its English meaning and record several short short pronunciation video clips. This helps to build a growing dataset that may modify and improve the performance of the AI recognition system in the future. Once the new data is saved, the vocabulary library will automatically update to display the newly added words, making the platform continuously expandable.

Moreover, the users can interact with an AI chatbot called “MySemai AI” which is accessible from the homepage of the platform. The AI chatbot is powered by Google Gemini model by integrating Gemini API and encourages users to ask questions about Bahasa Semai vocabulary and learning. This function offers additional guidance and explanation for the users, making the learning experience more interactive and supportive.

All in all, although the prototype achieved the main objectives of the project, further testing and development would be needed to improve the accuracy of the lip-reading recognition system and expand the vocabulary database. Nevertheless, the current prototype successfully demonstrates how artificial intelligence and web technologies can be used to support indigenous language preservation such as Bahasa Semai.

5.0 Prototype Showcase

5.1 Landing Page



Lip Read My Library About

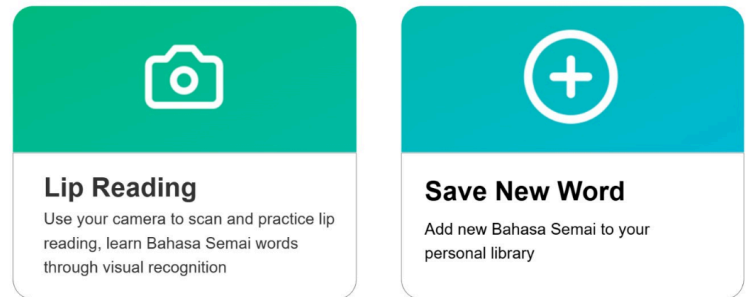


Figure 5.1: Landing page of the Bahasa Semai learning platform.

The page introduces the purpose of the website and provides navigation to the main features such as Lip Reading, My Library, Save New Word, and MySemai AI.

5.2 AI Lip Reading Feature

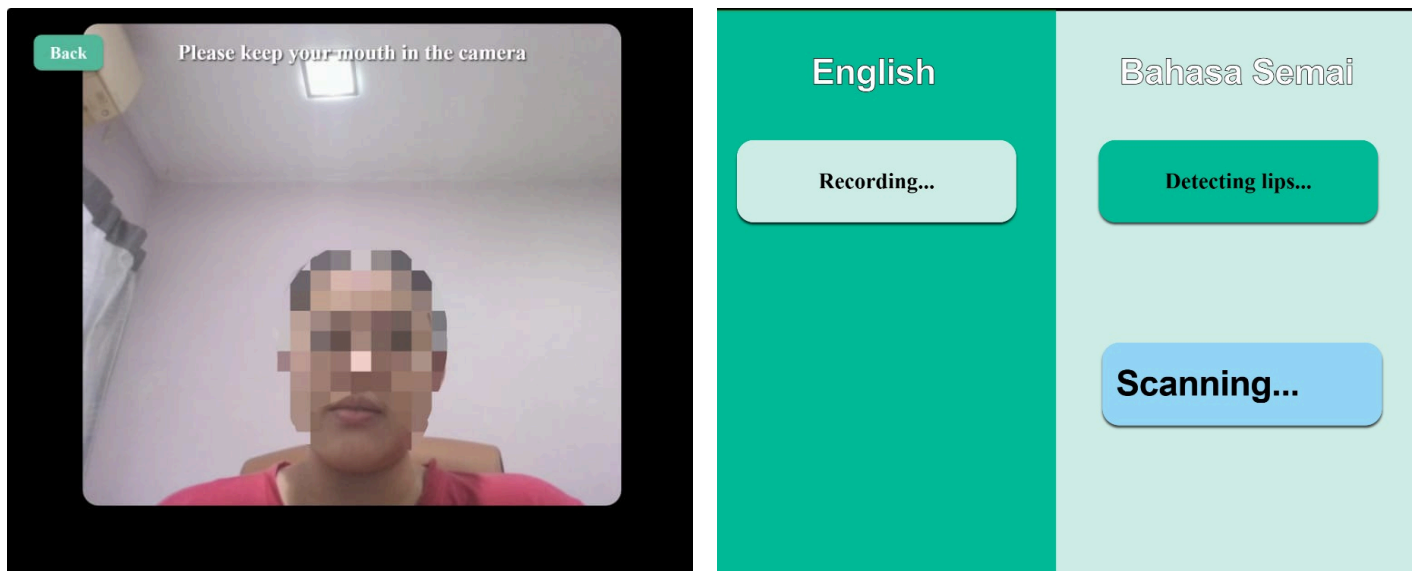


Figure 5.2: AI lip-reading feature that captures the user's lip movements through the camera to identify the spoken Bahasa Semai word.

5.3 Lip Reading Recognition Result

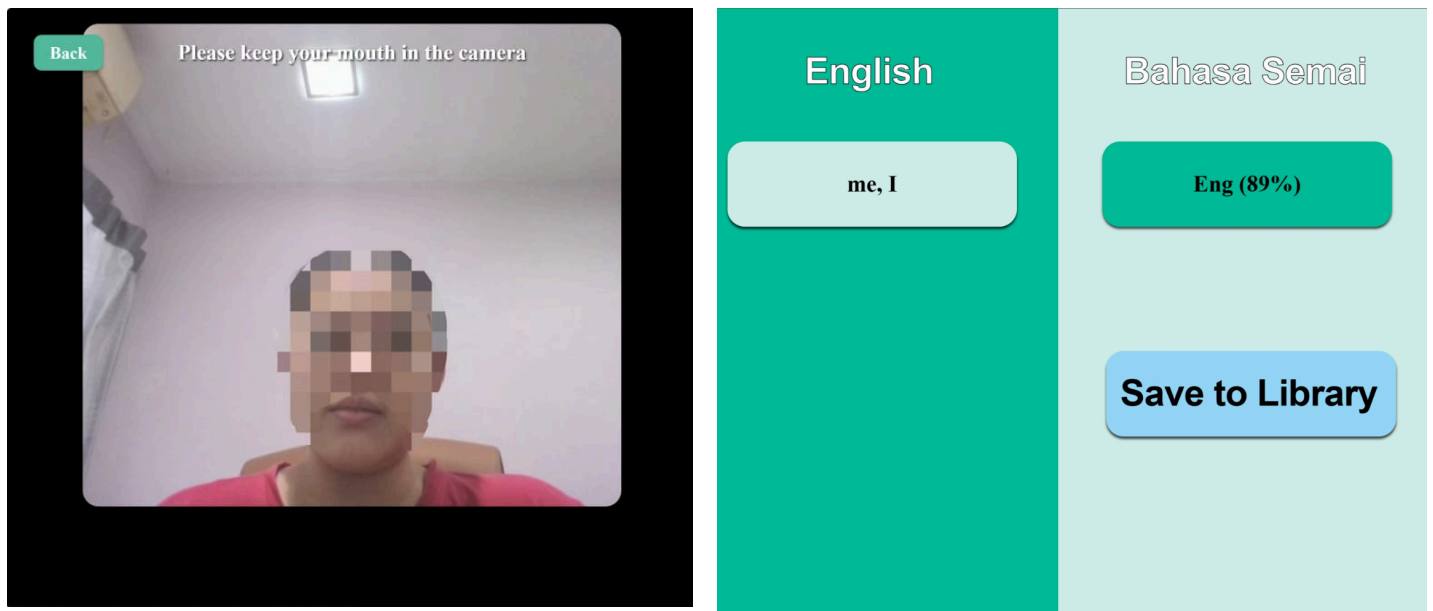
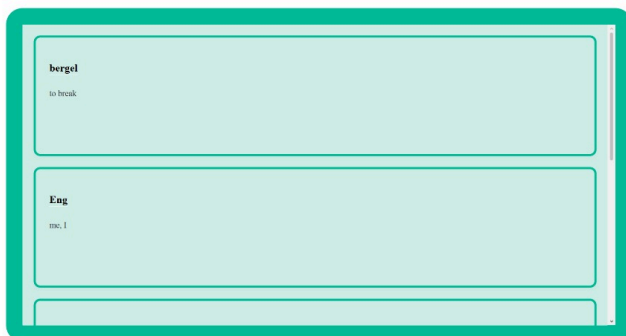


Figure 5.3: Example of the lip-reading recognition result showing the detected Bahasa Semai word and its English meaning with the confidence score generated by the AI system.

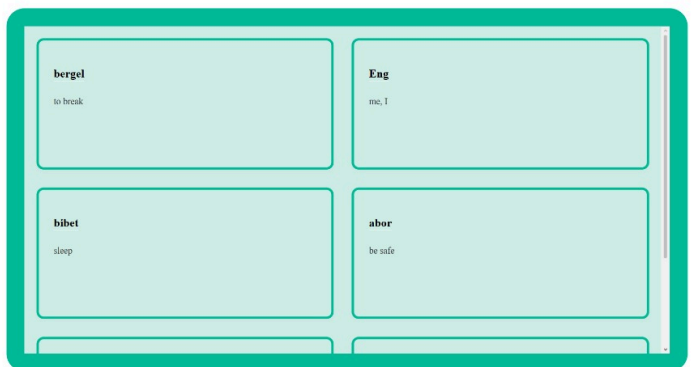
5.4 Vocabulary Library

Lip Read My Library About

Recent



Library



Click for more

Figure 5.4: Digital vocabulary library that displays Bahasa Semai words stored in the system database together with their meanings.

5.5 Save New Word Feature

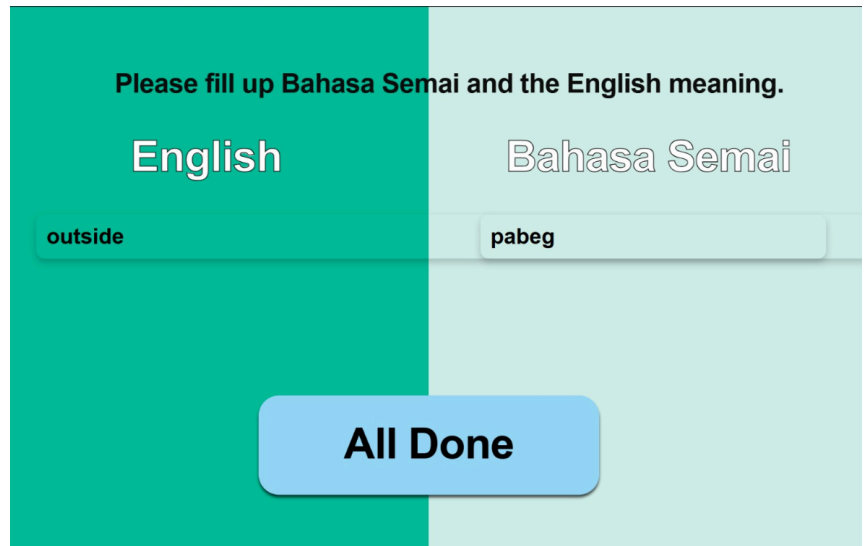
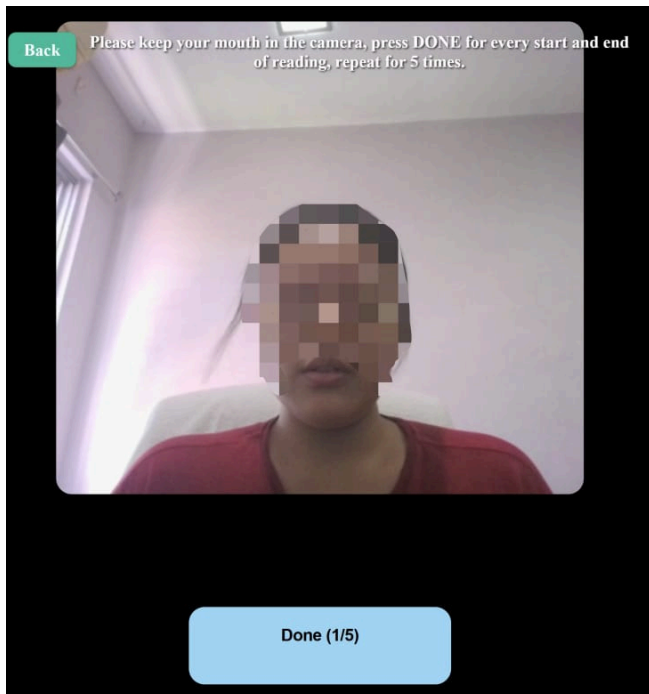


Figure 5.5: Word contribution page where users can add new Bahasa Semai vocabulary and record pronunciation video clips to expand the language dataset

5.6 MySemai AI Chatbot

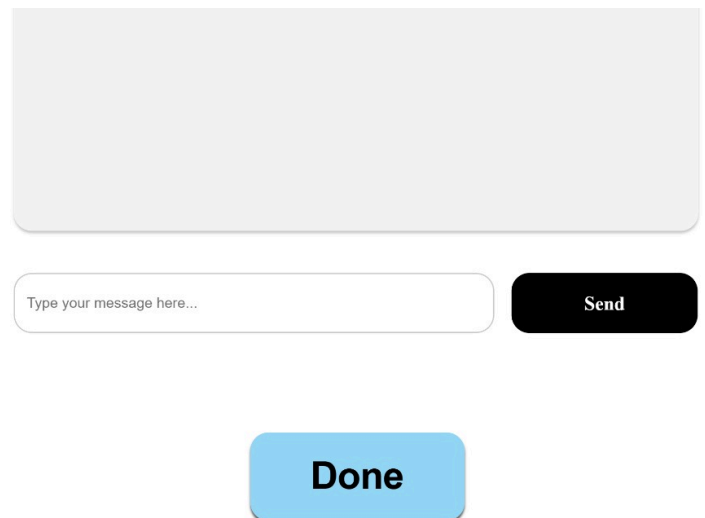
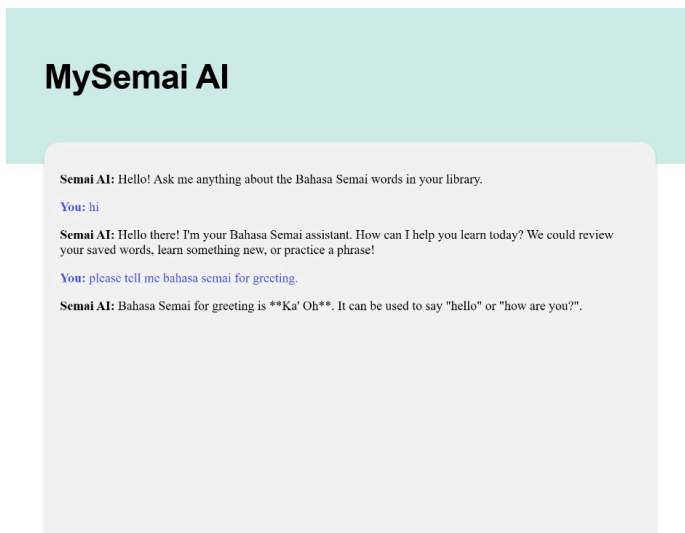


Figure 5.6: “MySemai AI” chatbot that allows users to ask questions about Bahasa Semai vocabulary and receive instant responses.

5.7 Cultural Information Page

Lip Read My Library About



Welcome to the World of Kaum Semai!

In the beautiful country of Malaysia, there is a very special and respected family of people called the Orang Asli. "Orang Asli" means the "Original People" or the "First People." The Semai people, or Kaum Semai, are one of the biggest and most peaceful groups in this wonderful family.

Tap a question to start exploring!

What is Bahasa Semai?

Where do Semai people live?

How do Semai people live?

Lip Read My Library About



Welcome to the World of Kaum Semai!

In the beautiful country of Malaysia, there is a very special and respected family of people called the Orang Asli. "Orang Asli" means the "Original People" or the "First People." The Semai people, or Kaum Semai, are one of the biggest and most peaceful groups in this wonderful family.

Tap a question to start exploring!

What is Bahasa Semai?



Just like you might speak English or Malay with your family, the Semai people have their very own beautiful language called Bahasa Semai. For many, many generations, grandparents and parents have taught this language to their children by telling wonderful stories about the forest, singing songs, and talking together.

Every language has its own special sounds and shapes. Today, by looking closely at how the mouth moves, we can learn to understand and speak these beautiful words together. By learning Bahasa Semai, we are showing respect to the First People of Malaysia and helping to keep their amazing language shining bright!



Where do Semai people live?

Many Semai families live close to the beautiful green rainforests, clear rivers, and tall mountains in places like Perak and Pahang.

They have a very deep love and respect for nature. For hundreds of years, they have been experts at living in harmony with the forest, growing their own food, and crafting beautiful things from bamboo and leaves.



How do Semai people live?

The Semai people are famous for being incredibly kind and peaceful, treating everyone in their village like one big, happy family. They have a beautiful tradition of always sharing what they have—like delicious fruits gathered from the forest—so no one is ever left out.

They love living closely with nature, building cozy homes from bamboo and leaves while always protecting the jungle. Instead of arguing when there is a problem, they sit down and talk gently until everyone is smiling and happy again.

Figure 5.7: Cultural information section providing educational content about the Semai community and the importance of preserving indigenous languages.

6.0 AI Acknowledgement

This project uses artificial intelligence tools to assist in the development and preparation of the report and project materials. The AI tools used in this project include ChatCPT and Google Gemini.

ChatGPT was used to assist in generating ideas, modifying sentence structure, and refining the written content of the report. It was also used to help organize sections such as the introduction, methodology, and literature review to ensure the report is clear and well-structured. These AI tools were used only as a supporting resource and references, while all final decisions, edits, and project concepts were reviewed and finalized by the project team.

On the other hand, Google Gemini was used in both the development component of the project and as a reference tool. In the project implementation, Gemini API was integrated into the platform to power the MySemai AI chatbot. This allows the users to question and have further understanding about Bahasa Semai by receiving instant AI-generated responses.

These AI tools were used only as supporting resources during the development and writing process. All design concepts, problem analysis, implementation decisions and final report content were carefully reviewed and finalized by the project team to ensure originality and relevance of the proposed solution.

7.0 Conclusion

In conclusion, an AI-powered web platform was designed to support the preservation and learning of Bahasa Semai. Bahasa Semai as one of the indigenous languages spoken by Semai people in Malaysia faces several challenges such as limited digital resources, lack of education platforms and declining usage among younger generations. These obstacles highlight the importance of developing digital approaches that can help preserve the language while making it easier for communities to learn and access.

This proposed platform emphasizes on building a digital language library where users can explore Bahasa Semai vocabulary along with their meanings and pronunciation examples. Moreover, one of the key features of the platform is the interactive pronunciation practice function. The lip-reading technology is introduced to analyze the user's mouth movements through the camera and provide an immediate feedback on pronunciation accuracy. This function lets users practice speaking the language efficiently instead of only reading or listening to it. In addition, this platform also encourages the users to contribute new words and pronunciation recordings which helps in expanding the language database over time. This combination of

language documentation and interactive learning contribute a more engaging learning experience and encourage users to practice speaking the language actively.

Additionally, the platform also includes an AI chatbot called “MySemai AI” which is powered by Google Gemini. It acts as an assistant by answering user’s questions related to Bahasa Semai vocabulary and learning. Therefore, users will obtain additional support and experience a more interactive and engaging environment from the platform.

Furthermore, this project aligns with the goals of the United Nations Sustainable Development Goal 4: Quality Education by offering accessible education resources that promote inclusive tuition and cultural preservation. By using modern technologies such as artificial intelligence, this platform indicates how digital tools can protect and promote indigenous languages in the modern digital era.

In the future, this project can be further improved by expanding and updating the language database with more vocabulary, phrases, and cultural stories that are contributed by native speakers. Moreover, additional features such as speech recognition technology, gamified learning activities and a mobile application version could also be designed to make the platform more interactive and accessible. With the help of community participation and continuous development, the platform has the opportunity to become a valuable digital resource for preserving Bahasa Semai and encouraging more people to learn and appreciate indigenous languages.

Reference

Department of Orang Asli Development. (n.d.). *Orang Asli Profile and Information*. Retrieved from <https://www.jakoa.gov.my>

Ethnologue. (2023). *Semai Language*. Retrieved from <https://www.ethnologue.com>

Google DeepMind. (2024). *Gemini: A family of multimodal AI models*. Retrieved from <https://deepmind.google/technologies/gemini/>

Google Research. (2023). *AI and Language Preservation*. Retrieved from <https://ai.google>

Lane, I., Waibel, A., & Black, A. (2019). *Speech and Language Technologies for Low-Resource Languages*. IEEE Signal Processing Magazine, 36(1), 53–61.

Prahalad, K., Black, A., & Mosur, R. (2012). *Subword Modeling for Speech Recognition in Low-Resource Languages*. IEEE Workshop on Spoken Language Technology.

UNESCO. (n.d.). *World Atlas of Languages*. Retrieved from <https://www.unesco.org/en/world-atlas-languages>

United Nations. (2015). *Sustainable Development Goals: Goal 4 – Quality Education*. Retrieved from <https://sdgs.un.org/goals/goal4>